

Keshav Krishna

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Summary

Data engineer with a strong background in designing and maintaining data pipelines and workflows using Python, SQL, Spark, Hive, and Airflow. Developed early-morning and late-night content mixes serving 12 M daily users, optimizing data models and deploying services in Docker containers. Expert in ETL, data modeling, and containerized deployments, seeking a summer internship to support investment research and analytics.

EDUCATION

New York University Courant

MS, Computer Science

- **GPA:** 4.0
- **Coursework:** Machine Learning, Honors Analysis of Algorithms, Programming Languages

Aug 2025 - May 2027

New York City

Indian Institute of Technology, Ropar

Bachelors, Computer Science

- **GPA:** 8.74/10
- **Achievements:** Merit Scholarship for top performance

Aug 2019 - Jun 2023

Ropar, India

SKILLS SUMMARY

- **Programming Languages:** C, C++, Python, SQL, Unix scripting
- **Data Engineering & Architecture:** ETL, Airflow, pyspark, hive, Hadoop, Data Wrangling, Data Architectures, Data Modeling, Database Design, databases, NoSQL
- **Databases & Storage:** Postgres, MYSQL, redis, Qdrant, Annoy
- **Analytics & Machine Learning:** Machine learning, deep learning, LLMs, Tensorflow, Hugging Face, Pytorch, Analytical Skills, PowerBI
- **Cloud, DevOps & Tools:** Docker, Kubernetes, AWS, GIT, JIRA, FastAPI, backend, Agile

EXPERIENCE

JioSaavn

Software Engineer(Data Science)

- Orchestrated ETL pipelines using Hive, Spark, and Airflow on AWS to build early-morning and late-night content mixes for 12M daily users with Redis and PHP, driving 1M impressions, 800K clicks, and 500K streams per day
- Generated mood-based topic mixes for 7.6M daily users using Hive, Spark, and TF-IDF, implemented data modeling and deduplication filters and Flask endpoints in the PHP backend, and served to 100M app users, achieving a 12.81% CTR and 8.5 streams per user
- Assessed few-shot SetFit and transformer models (hing-roberta, muril-base) and transitioned to prompt engineering with Llama3-8B-instruct, achieving 85% accuracy on manually labeled data, thus improving the content moderation of user-generated playlists
- Created a custom BERT model to generate song and user embeddings, which raised top-5 and top-10 accuracy to 61.82% and 67.21%, thereby improving user-learning capabilities and recommendation personalization
- Developed a RAG system using LLaMA3-7B for prompt intent classification, dynamically retrieving songs to assemble playlists and generating custom cover art via the segmid/SSD-1B diffusion model, enhancing user interaction and playlist customization
- Reduced infrastructure costs by migrating data from Annoy indices to Qdrant, purging unused data, and cleaning tables within a modular data architecture for improved lifecycle management

Jul 2023 - Jul 2025

Mumbai, India

GE Healthcare

Software Intern

- Applied YOLO algorithm to train a deep-learning model that identified and blurred faces in medical images, leveraging Python data wrangling scripts to process over 20,000 images and achieving over 90% accuracy
- Trained YOLO models to detect people in images(IR and RGB), reaching over 85% mean average precision and accuracy
- Created Docker images and deployed containerized FastAPI modules on Kubernetes clusters, simplifying import and utilization in various projects

May 2022 - Jul 2022

Bangalore, India

RESEARCH & ACADEMIC PROJECTS

Review of machine learning innovations in cache management

<https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1441250/abstract>

Jan 2024 - Apr 2024

Independent Research

- Single-author literature review of 40+ papers on ML for cache systems (replacement policies, dynamic management, edge/content caching, security).
- Evaluated methods (reinforcement learning, LSTM, imitation learning, deep neural nets) for performance optimization and attack detection; synthesized gaps and practical recommendations.
- Published in Frontiers in AI; provided roadmap for research and practitioner adoption with 9+ citations in few months.

Mumbai

Pico-LLM: Unified SLM Training & Interpretability | <https://github.com/k-keshav-k/pico-llm/tree/main>

NYU Courant

- **Unified Training Framework:** Developed a codebase for Small Language Models (SLMs) training from scratch along with comparison with lstm and kmeans.
- **Mathematical Reasoning:** Extended base model trained to also include reasoning by supporting **SFT** and **DPO** alignment on reasoning datasets like GSM8K.
- **Mechanistic Interpretability:** Implemented visualization tools (Attention Maps, Training loss, Test loss, T-SNE) to diagnose feature activations and internal model states.
- **Inference Optimization:** Engineered a robust inference engine with **KV-Caching** and automatic architecture detection for efficient batched generation.

Sep 2025 - Dec 2025

New York City, NY

ACHIEVEMENTS

- **Merit Scholarship:**IIT Ropar Merit Scholarship Cash scholarship provided to top 7% students by CGPA all over the batch(2019-20) link
- **Mathematics Olympiad:**Gold Medal at RMO Top Scorer at RMO (Regional Mathematics Olympiad) conducted by HBCSE (2017)
- **Hackathon Win:**Jumpstart'22 Finalist Hackathon conducted by Publicis Sapient. Passed all rounds from among 29864 people(2022)
- **Competitive Coding Win:**Competitive Coding Codechef 5 star, rating 2021, Codeforces Specialist (2022)
- **Google competition Top performance:**Google Kickstart Round D Got Global rank 1547 out of 7500+ registrations (2022)
- **IIT Entrance exam top rank:**JEE Mains & Advanced Mains - Got AIR 2505 out of 1.14 million & Advanced - Got AIR 3293 out 161,000 (2019)